KESTREL: An Offline, On-Board AI System for Autonomous Aerial Search and Rescue

Team KESTREL

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Source code

1. Motivation

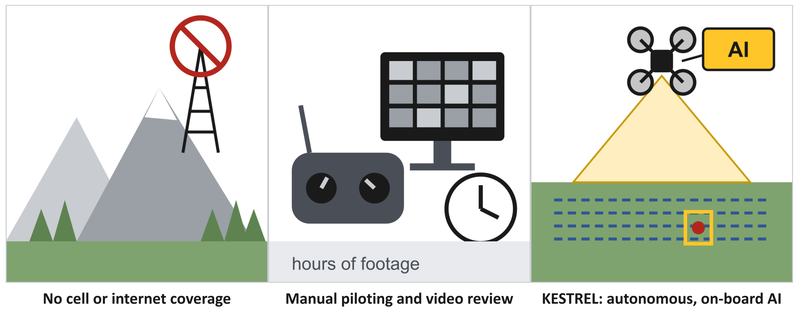


Figure 1. Search and rescue challenges and the proposed approach (schematic).

2. System Overview

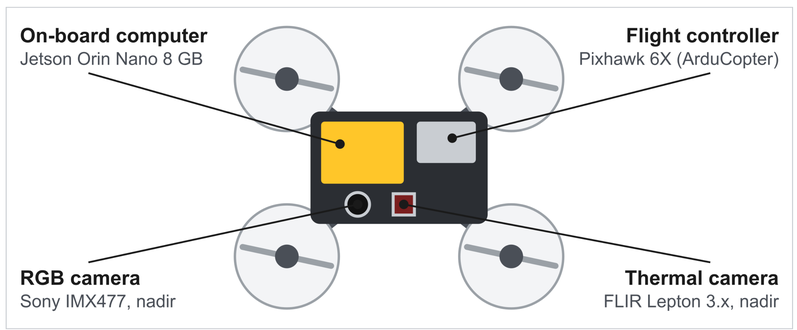


Figure 2. Airframe layout (top view, schematic).

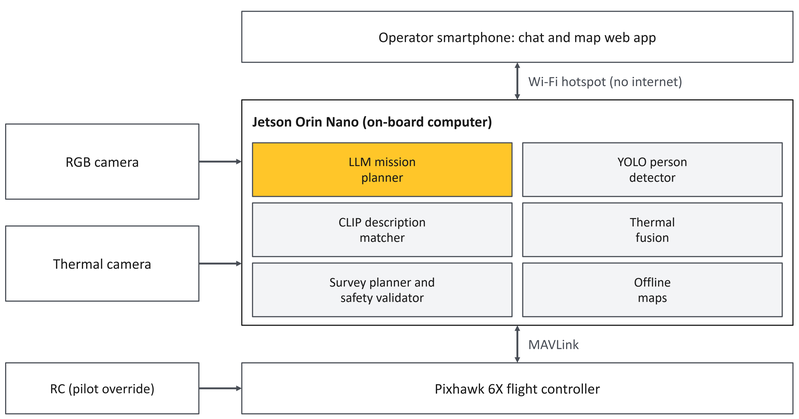


Figure 3. System architecture. All inference runs on the aircraft.

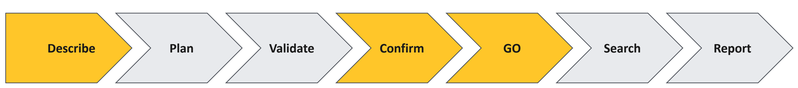


Figure 4. Operational workflow. Gold steps require a human; grey steps are automated.

3. Methods

3.1 Mission planning by conversation

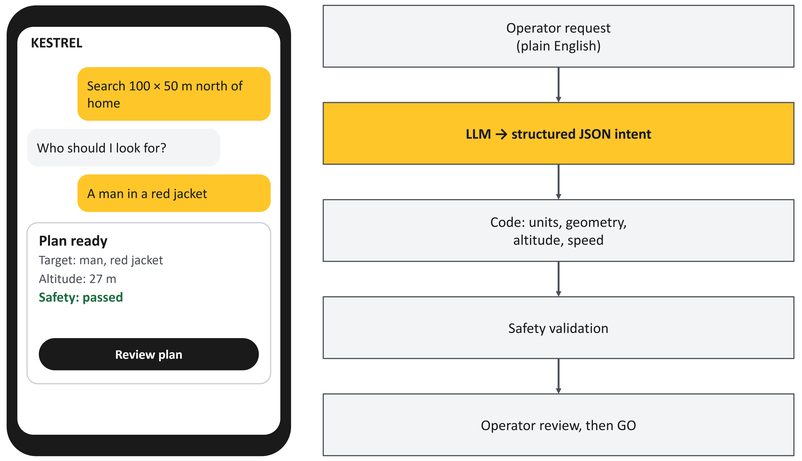


Figure 5. The operator chats in plain English. The LLM only interprets the words; code computes every number and never lets the model produce coordinates.

3.2 Coverage path and altitude

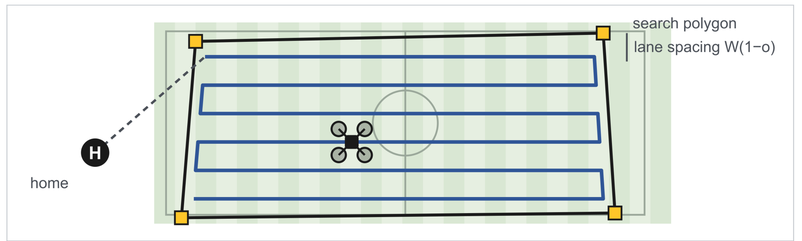


Figure 6. Lawn-mower lanes cover the search area; each point is seen in at least 3 frames.

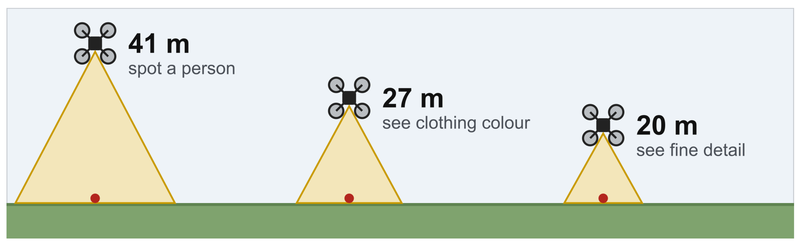
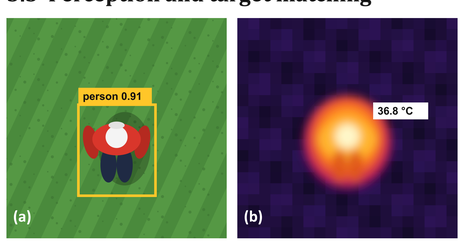


Figure 7. Altitude is chosen automatically from the detail the description needs (80 / 120 / 160 px on target).

3.3 Perception and target matching



Thermal fusion
$C_{person} = 1 - (1 - C_{yolo})(1 - w \cdot C_{heat})$

CLIP description match
$C_{final} = C_{person} \cdot C_{match}$

Alert
mean of last 5 $C_{final} \geq 0.7$

Figure 8. (a) RGB person detection and (b) radiometric thermal view (schematic), fused and matched to the description.

4. Safety and Regulatory Compliance

- ✓ GPS fix and home
- ✓ Battery level
- ✓ Geofence and failsafes
- ✓ 400 ft / 120 m ceiling
- ✓ LAANC altitude limit
- ✓ RC link to pilot
- ✓ Autopilot pre-arm
- ✓ Mission read-back
- ✓ Cameras and AI ready
- ✓ Legal checklist

Figure 9. Automated checks run at upload and again at GO; any failure blocks launch.

Part 107. Flown by a certified Remote Pilot in Command with RC override; the operator needs no piloting skills.

5. Results

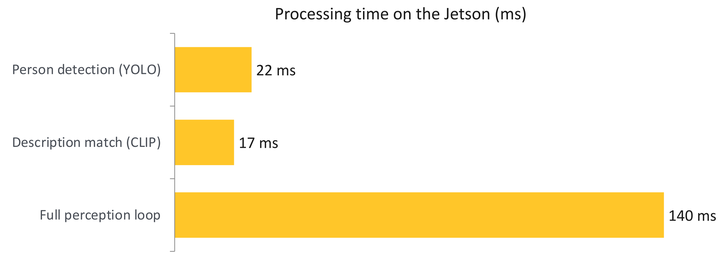


Figure 10. Measured latency; the full loop uses about 14 % of its 1 s cycle.

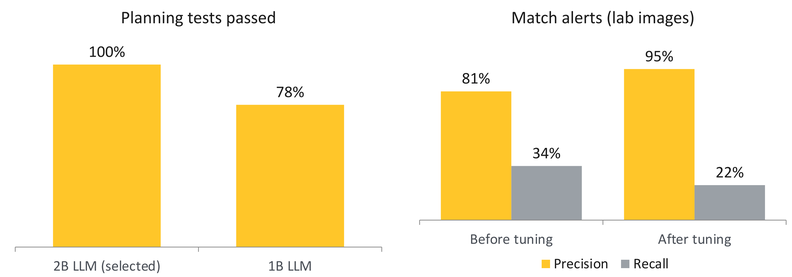


Figure 11. Left: LLM choice on 16 live planning tests. Right: CLIP threshold tuning trades recall for precision.

631	0	8 GB
automated tests	network connections	total on-board memory

6. Conclusions

- The full search workflow runs offline on an 8 GB embedded computer.
- The LLM interprets language; code makes every safety-critical decision.
- Next: aerial validation and supervised field trials at KSU Marietta.

References. Ultralytics YOLO; OpenCLIP (Cherti et al., CVPR 2023); llama.cpp; ArduPilot; FAA 14 CFR Part 107.